

Linux Log File Analysis, Automation, and SIEM Visualisation

Cybersecurity Project

Objective 1: Manual Log Analysis

Learn how to analyse computer logs to identify suspicious login activity. This is a key skill for SOC (Security Operations Center) Analysts, who monitor systems for signs of attack.

Tools Required/Prerequisites

1. Basic knowledge of authentication and SSH
2. VS Code - viewing and editing log files
3. Excel/Google Sheets - organising and analysing extracted log data
4. Sample log file - https://github.com/logpai/loghub/blob/master/Linux/Linux_2k.log

Steps Taken:

Step 1: Download a Sample Log File

Step 2: Open and Review the Logs

Step 3: Identify Suspicious Events

Step 4: Organise Findings

Step 5: Summarise the Findings and Provide Recommendations

Objective 2: Automated Log Analysis

Extend the beginner-level manual log review by writing a simple Python script in VS Code to automatically scan Linux log files for suspicious activity (failed logins, unknown users, etc.). Focus on log lines 200–500 to simulate handling larger data more efficiently.

Tools Required/Prerequisites

1. Objective 1 completed
2. VS code installed
3. Python 3.x set up
4. Basic knowledge of running a Python script required
5. The Linux_2k.log file from LogHub is the same as the beginner project

Steps Taken:

Step 1: Open VS Code and Create a New Python File

Step 2: Write the Log Analysis Script

Step 3: Run the Script

Step 4: Export the Results

Objective 3: Log Analysis and Visualisation with Splunk

Building on the manual and automated log analysis, this objective introduces Splunk, a Security Information and Event Management (SIEM) tool used in professional SOC environments.

Tools Required/Prerequisites

1. Objective 1 and 2 completed
2. Access to a Splunk Enterprise
3. The Linux_2k.log file from LogHub
4. Basic understanding of searching and filtering data in Splunk

Steps Taken:

Step 1: Download and Set Up Splunk

Step 2: Upload the Log File

Step 3: Search and Filter Suspicious Activity

Step 4: Visualise the Data

Findings:

Objective 1: manual log analysis

a. Identify suspicious events from raw data → lines 1 to 40

```
1 Jun 14 15:16:01 combo sshd(pam_unix)[19939]: authentication failure; logname= uid=0 euid=0 tty=NODEVssh ruser= rhost=218.188.2.4
2 Jun 14 15:16:02 combo sshd(pam_unix)[19937]: check pass; user unknown
3 Jun 14 15:16:02 combo sshd(pam_unix)[19937]: authentication failure; logname= uid=0 euid=0 tty=NODEVssh ruser= rhost=218.188.2.4
4 Jun 15 02:04:59 combo sshd(pam_unix)[20882]: authentication failure; logname= uid=0 euid=0 tty=NODEVssh ruser= rhost=220-135-151-1.hinet-ip.hinet.net user=root
5 Jun 15 02:04:59 combo sshd(pam_unix)[20884]: authentication failure; logname= uid=0 euid=0 tty=NODEVssh ruser= rhost=220-135-151-1.hinet-ip.hinet.net user=root
6 Jun 15 02:04:59 combo sshd(pam_unix)[20883]: authentication failure; logname= uid=0 euid=0 tty=NODEVssh ruser= rhost=220-135-151-1.hinet-ip.hinet.net user=root
7 Jun 15 02:04:59 combo sshd(pam_unix)[20885]: authentication failure; logname= uid=0 euid=0 tty=NODEVssh ruser= rhost=220-135-151-1.hinet-ip.hinet.net user=root
8 Jun 15 02:04:59 combo sshd(pam_unix)[20886]: authentication failure; logname= uid=0 euid=0 tty=NODEVssh ruser= rhost=220-135-151-1.hinet-ip.hinet.net user=root
9 Jun 15 02:04:59 combo sshd(pam_unix)[20892]: authentication failure; logname= uid=0 euid=0 tty=NODEVssh ruser= rhost=220-135-151-1.hinet-ip.hinet.net user=root
10 Jun 15 02:04:59 combo sshd(pam_unix)[20893]: authentication failure; logname= uid=0 euid=0 tty=NODEVssh ruser= rhost=220-135-151-1.hinet-ip.hinet.net user=root
11 Jun 15 02:04:59 combo sshd(pam_unix)[20896]: authentication failure; logname= uid=0 euid=0 tty=NODEVssh ruser= rhost=220-135-151-1.hinet-ip.hinet.net user=root
12 Jun 15 02:04:59 combo sshd(pam_unix)[20897]: authentication failure; logname= uid=0 euid=0 tty=NODEVssh ruser= rhost=220-135-151-1.hinet-ip.hinet.net user=root
13 Jun 15 02:04:59 combo sshd(pam_unix)[20898]: authentication failure; logname= uid=0 euid=0 tty=NODEVssh ruser= rhost=220-135-151-1.hinet-ip.hinet.net user=root
14 Jun 15 04:06:18 combo su(pam_unix)[21416]: session opened for user cyrus by (uid=0)
15 Jun 15 04:06:19 combo su(pam_unix)[21416]: session closed for user cyrus
16 Jun 15 04:06:20 combo logrotate: ALERT exited abnormally with [1]
17 Jun 15 04:12:42 combo su(pam_unix)[22644]: session opened for user news by (uid=0)
18 Jun 15 04:12:43 combo su(pam_unix)[22644]: session closed for user news
19 Jun 15 12:12:34 combo sshd(pam_unix)[23397]: check pass; user unknown
20 Jun 15 12:12:34 combo sshd(pam_unix)[23397]: authentication failure; logname= uid=0 euid=0 tty=NODEVssh ruser= rhost=218.188.2.4
21 Jun 15 12:12:34 combo sshd(pam_unix)[23395]: check pass; user unknown
22 Jun 15 12:12:34 combo sshd(pam_unix)[23395]: authentication failure; logname= uid=0 euid=0 tty=NODEVssh ruser= rhost=218.188.2.4
23 Jun 15 12:12:34 combo sshd(pam_unix)[23404]: check pass; user unknown
24 Jun 15 12:12:34 combo sshd(pam_unix)[23404]: authentication failure; logname= uid=0 euid=0 tty=NODEVssh ruser= rhost=218.188.2.4
25 Jun 15 12:12:34 combo sshd(pam_unix)[23399]: check pass; user unknown
26 Jun 15 12:12:34 combo sshd(pam_unix)[23399]: authentication failure; logname= uid=0 euid=0 tty=NODEVssh ruser= rhost=218.188.2.4
27 Jun 15 12:12:34 combo sshd(pam_unix)[23406]: check pass; user unknown
28 Jun 15 12:12:34 combo sshd(pam_unix)[23406]: authentication failure; logname= uid=0 euid=0 tty=NODEVssh ruser= rhost=218.188.2.4
29 Jun 15 12:12:34 combo sshd(pam_unix)[23396]: check pass; user unknown
30 Jun 15 12:12:34 combo sshd(pam_unix)[23394]: check pass; user unknown
31 Jun 15 12:12:34 combo sshd(pam_unix)[23407]: check pass; user unknown
32 Jun 15 12:12:34 combo sshd(pam_unix)[23394]: authentication failure; logname= uid=0 euid=0 tty=NODEVssh ruser= rhost=218.188.2.4
33 Jun 15 12:12:34 combo sshd(pam_unix)[23403]: check pass; user unknown
34 Jun 15 12:12:34 combo sshd(pam_unix)[23396]: authentication failure; logname= uid=0 euid=0 tty=NODEVssh ruser= rhost=218.188.2.4
35 Jun 15 12:12:34 combo sshd(pam_unix)[23407]: authentication failure; logname= uid=0 euid=0 tty=NODEVssh ruser= rhost=218.188.2.4
36 Jun 15 12:12:34 combo sshd(pam_unix)[23403]: authentication failure; logname= uid=0 euid=0 tty=NODEVssh ruser= rhost=218.188.2.4
37 Jun 15 12:12:34 combo sshd(pam_unix)[23412]: check pass; user unknown
38 Jun 15 12:12:34 combo sshd(pam_unix)[23412]: authentication failure; logname= uid=0 euid=0 tty=NODEVssh ruser= rhost=218.188.2.4
39 Jun 15 12:13:19 combo sshd(pam_unix)[23414]: check pass; user unknown
40 Jun 15 12:13:19 combo sshd(pam_unix)[23414]: authentication failure; logname= uid=0 euid=0 tty=NODEVssh ruser= rhost=218.188.2.4
```

b. Tubulate findings

Timestamps	Source IP/Hostname	Action Type	Target User	Outcome	Notes
Jun 14 15:16:01	218.188.2.4	ssh authentication attempt	unknown	authentication failure	unknown user

Jun 14 15:16:02	218.188.2.4	ssh authentication attempt	unknown	user unknown/fail ure	unknown user
Jun 15 2:04:59	220-135-151-1.hinet- ip.hinet.net	ssh authentication attempt	root	authenticatio n failure	10 attempts - brute-force
Jun 15 4:06:18	localhost	su session	cyrus	session opened	system activity
Jun 15 4:06:19	localhost	su session	cyrus	session closed	system activity
Jun 15 4:06:20	localhost	logrotate alert	-	error (exit code 1)	error - misconfig or missing file
Jun 15 4:12:42	localhost	su session	news	session opened	system activity
Jun 15 4:12:43	localhost	su session	news	session closed	system activity
Jun 15 12:12:34	218.188.2.4	ssh authentication attempt	unknown	user unknown/fail ure	10 attempts - brute-force
Jun 15 12:13:19	218.188.2.4	ssh authentication attempt	unknown	user unknown/fail ure	brute-force

c. Summary of Findings

Attack Type: The log data indicates a high-volume, automated SSH brute-force attack conducted over a short timeframe.

Attack Vector: Multiple authentication requests were directed at both valid system accounts (e.g., root) and unknown usernames, indicating an unguided dictionary attack.

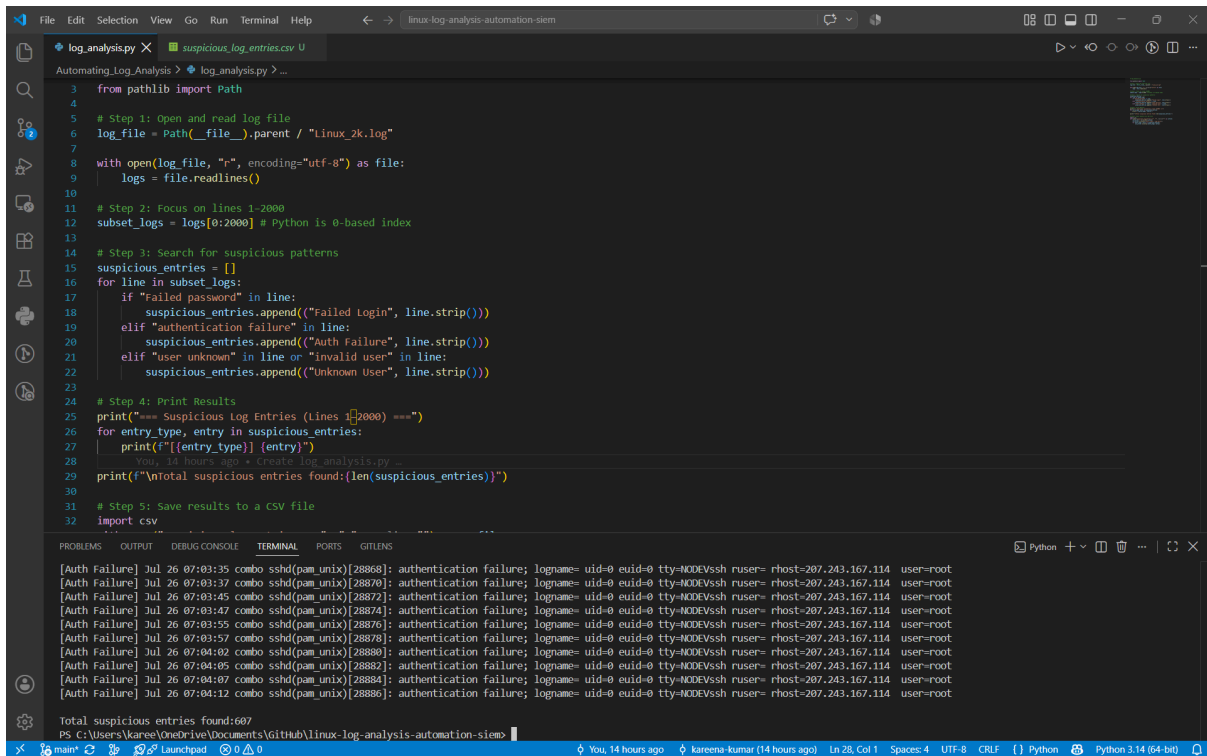
Indicators of Compromise (IoCs): Two primary external sources were identified as responsible for the suspicious failed login attempts:

- Source IP: 218.188.2.4
- Hostname: 220-135-151-1.hinet-ip.hinet.net

Recommendations for Remediation

- **Firewall Rules:** Immediately block all incoming traffic from both malicious IoCs (218.188.2.4 and 220-135-151-1.hinet-ip.hinet.net) at the perimeter firewall or host level.
- **Rate Limiting & Banning:** Deploy an automated intrusion prevention tool to enforce rate limiting and automatically ban IP addresses after 3–5 consecutive failed login attempts.

Objective 2: automated log analysis



The screenshot shows a VS Code editor with a Python script named `log_analysis.py` and its terminal output. The script automates log analysis by opening a log file, searching for suspicious patterns (failed password, authentication failure, invalid user, unknown user), and saving the results to a CSV file. The terminal output shows the results of the analysis, including a list of suspicious log entries and the total count of suspicious entries found (607).

```
3 from pathlib import Path
4
5 # Step 1: Open and read log file
6 log_file = Path(__file__).parent / "Linux_2k.log"
7
8 with open(log_file, "r", encoding="utf-8") as file:
9     logs = file.readlines()
10
11 # Step 2: Focus on lines 1-2000
12 subset_logs = logs[0:2000] # Python is 0-based index
13
14 # Step 3: Search for suspicious patterns
15 suspicious_entries = []
16 for line in subset_logs:
17     if "failed password" in line:
18         suspicious_entries.append(("Failed Login", line.strip()))
19     elif "authentication failure" in line:
20         suspicious_entries.append(("Auth Failure", line.strip()))
21     elif "user unknown" in line or "invalid user" in line:
22         suspicious_entries.append(("Unknown User", line.strip()))
23
24 # Step 4: Print Results
25 print("==== Suspicious Log Entries (Lines 1-2000) ====")
26 for entry_type, entry in suspicious_entries:
27     print(f"[{entry_type}] {entry}")
28
29 print(f"\nTotal suspicious entries found: {len(suspicious_entries)}")
30
31 # Step 5: Save results to a CSV file
32 import csv
```

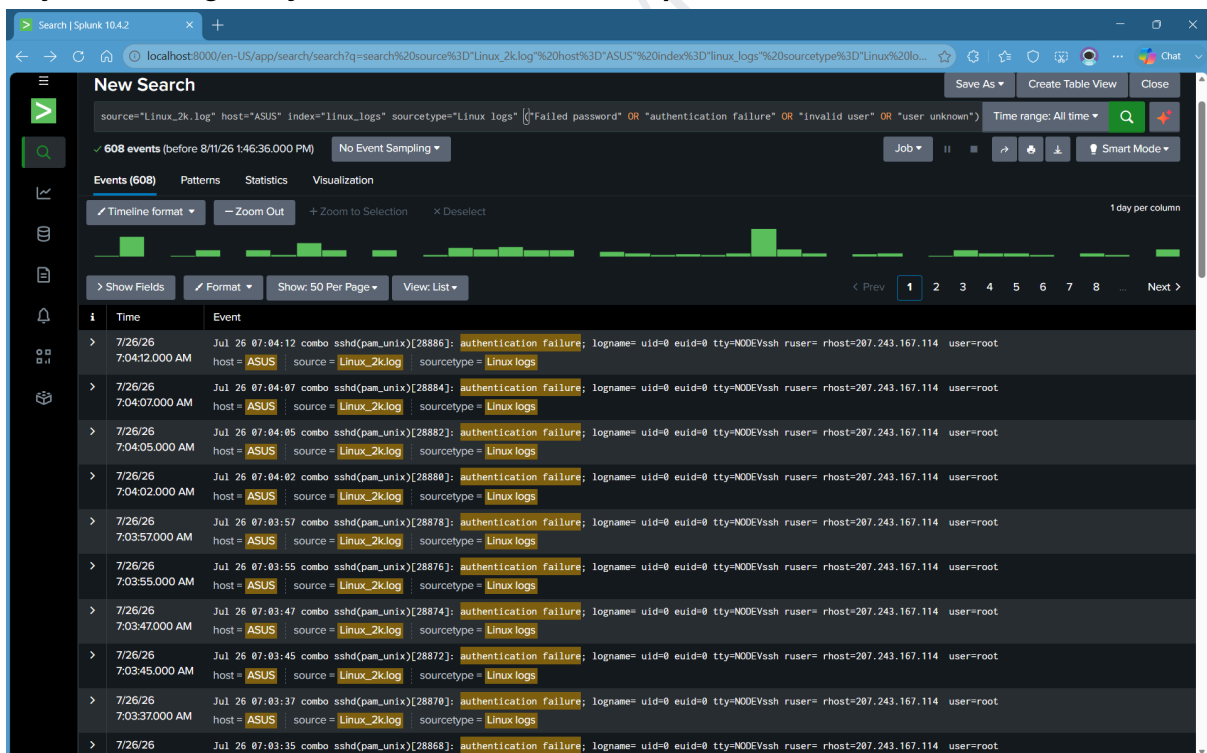
Terminal Output:

```
[Auth Failure] Jul 26 07:03:35 combo sshd(pam_unix)[28868]: authentication failure; logname=uid=0 euid=0 tty=NODEVssh ruser= rhost=207.243.167.114 user=root
[Auth Failure] Jul 26 07:03:37 combo sshd(pam_unix)[28870]: authentication failure; logname=uid=0 euid=0 tty=NODEVssh ruser= rhost=207.243.167.114 user=root
[Auth Failure] Jul 26 07:03:45 combo sshd(pam_unix)[28872]: authentication failure; logname=uid=0 euid=0 tty=NODEVssh ruser= rhost=207.243.167.114 user=root
[Auth Failure] Jul 26 07:03:47 combo sshd(pam_unix)[28874]: authentication failure; logname=uid=0 euid=0 tty=NODEVssh ruser= rhost=207.243.167.114 user=root
[Auth Failure] Jul 26 07:03:55 combo sshd(pam_unix)[28876]: authentication failure; logname=uid=0 euid=0 tty=NODEVssh ruser= rhost=207.243.167.114 user=root
[Auth Failure] Jul 26 07:03:57 combo sshd(pam_unix)[28878]: authentication failure; logname=uid=0 euid=0 tty=NODEVssh ruser= rhost=207.243.167.114 user=root
[Auth Failure] Jul 26 07:04:02 combo sshd(pam_unix)[28880]: authentication failure; logname=uid=0 euid=0 tty=NODEVssh ruser= rhost=207.243.167.114 user=root
[Auth Failure] Jul 26 07:04:05 combo sshd(pam_unix)[28882]: authentication failure; logname=uid=0 euid=0 tty=NODEVssh ruser= rhost=207.243.167.114 user=root
[Auth Failure] Jul 26 07:04:07 combo sshd(pam_unix)[28884]: authentication failure; logname=uid=0 euid=0 tty=NODEVssh ruser= rhost=207.243.167.114 user=root
[Auth Failure] Jul 26 07:04:12 combo sshd(pam_unix)[28886]: authentication failure; logname=uid=0 euid=0 tty=NODEVssh ruser= rhost=207.243.167.114 user=root

Total suspicious entries found: 607
PS C:\Users\karee\OneDrive\Documents\GitHub\linux-log-analysis-automation-siem>
```

607 suspicious entries found on VS code

Objective 3: log analysis and visualisation on Splunk



Keyword search of "Failed password" OR "authentication failure" OR "invalid user" OR "user unknown" resulted in 607 event results.

The screenshot shows a Splunk search interface with the following search query: `index="linux_logs" source="linux_2k.log" sourcetype="Linux logs" ("Failed password" OR "authentication failure" OR "invalid user" OR "user unknown") "user=guest"`. The results show 17 events from June 23, 2023, at 01:41:30 AM. Each event is an authentication failure for the user 'guest' from the host 'ASUS' via 'Linux_2klog'.

i	Time	Event
>	6/23/26 14:13:20.000 AM	Jun 23 01:41:32 combo sshd(pam_unix)[18967]: authentication failure; logname=uid=0 euid=0 tty=NODEVssh ruser= rhost=209.152.168.249 user=guest host = ASUS source = Linux_2klog sourcetype = Linux logs
>	6/23/26 14:13:00.000 AM	Jun 23 01:41:30 combo sshd(pam_unix)[18976]: authentication failure; logname=uid=0 euid=0 tty=NODEVssh ruser= rhost=209.152.168.249 user=guest host = ASUS source = Linux_2klog sourcetype = Linux logs
>	6/23/26 14:13:00.000 AM	Jun 23 01:41:30 combo sshd(pam_unix)[18981]: authentication failure; logname=uid=0 euid=0 tty=NODEVssh ruser= rhost=209.152.168.249 user=guest host = ASUS source = Linux_2klog sourcetype = Linux logs
>	6/23/26 14:13:00.000 AM	Jun 23 01:41:30 combo sshd(pam_unix)[18982]: authentication failure; logname=uid=0 euid=0 tty=NODEVssh ruser= rhost=209.152.168.249 user=guest host = ASUS source = Linux_2klog sourcetype = Linux logs
>	6/23/26 14:13:00.000 AM	Jun 23 01:41:30 combo sshd(pam_unix)[18983]: authentication failure; logname=uid=0 euid=0 tty=NODEVssh ruser= rhost=209.152.168.249 user=guest host = ASUS source = Linux_2klog sourcetype = Linux logs
>	6/23/26 14:12:30.000 AM	Jun 23 01:41:29 combo sshd(pam_unix)[18977]: authentication failure; logname=uid=0 euid=0 tty=NODEVssh ruser= rhost=209.152.168.249 user=guest host = ASUS source = Linux_2klog sourcetype = Linux logs
>	6/23/26 14:12:30.000 AM	Jun 23 01:41:29 combo sshd(pam_unix)[18975]: authentication failure; logname=uid=0 euid=0 tty=NODEVssh ruser= rhost=209.152.168.249 user=guest host = ASUS source = Linux_2klog sourcetype = Linux logs
>	6/23/26 14:12:30.000 AM	Jun 23 01:41:29 combo sshd(pam_unix)[18973]: authentication failure; logname=uid=0 euid=0 tty=NODEVssh ruser= rhost=209.152.168.249 user=guest host = ASUS source = Linux_2klog sourcetype = Linux logs
>	6/23/26 14:12:30.000 AM	Jun 23 01:41:29 combo sshd(pam_unix)[18971]: authentication failure; logname=uid=0 euid=0 tty=NODEVssh ruser= rhost=209.152.168.249 user=guest host = ASUS source = Linux_2klog sourcetype = Linux logs

"user=guest" resulted in 17 events

src/ip=209.152.168.249, 10 events between Jun 23 01:41:29 and Jun 23 01:41:32

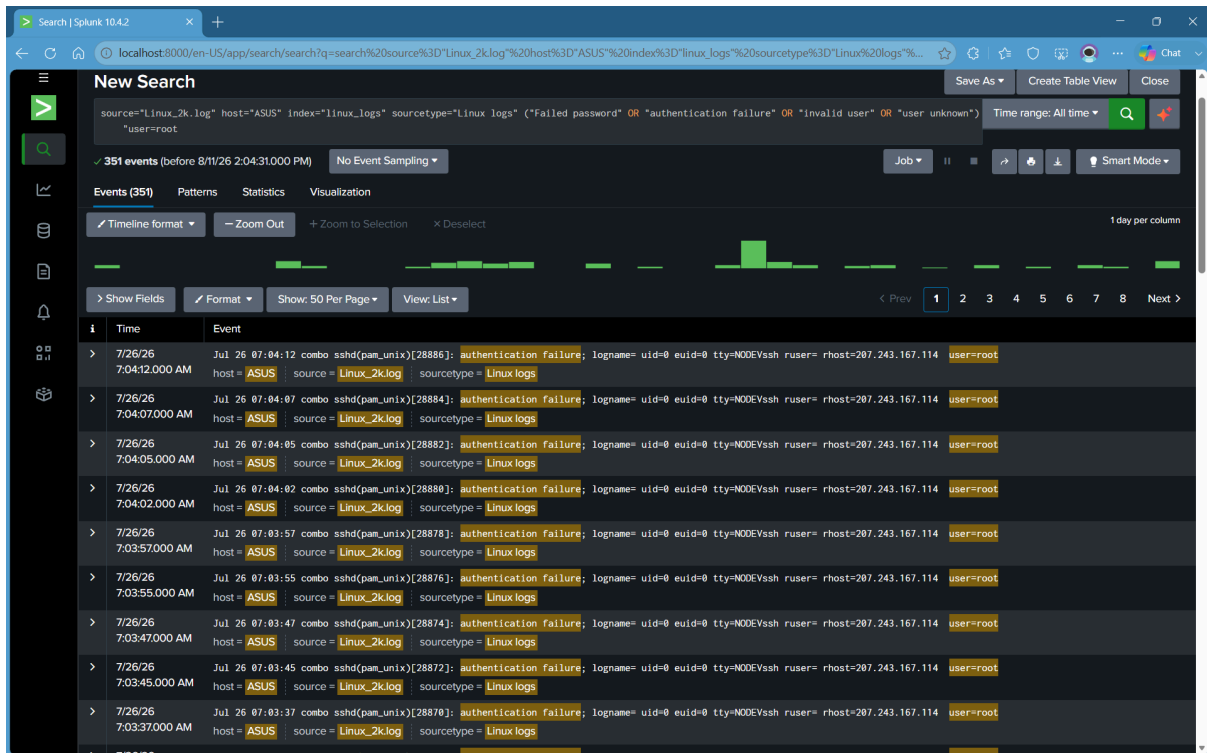
src/ip=217.60.212.66, 6 events between Jun 21 08:56:36 and Jun 21 08:56:36

src/ip=211.46.224.253, 1 events on Jun 17 19:43:13

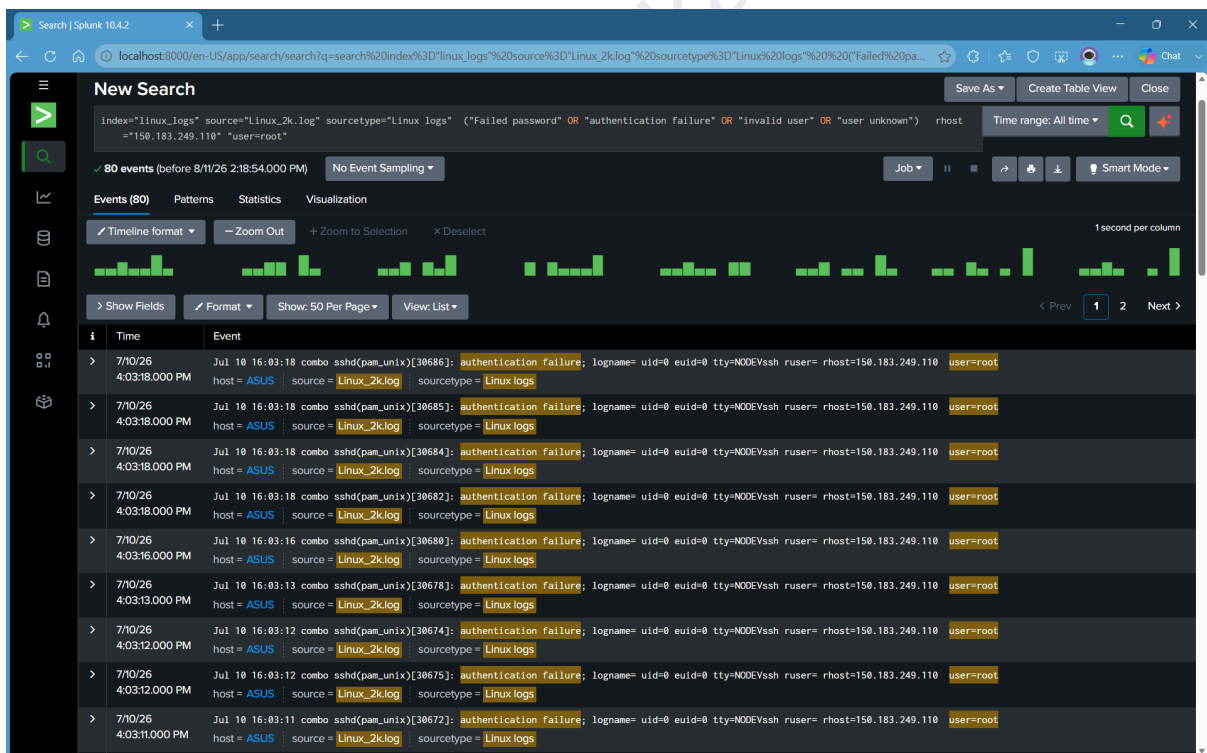
The screenshot shows a Splunk search interface with the following search query: `index="linux_logs" source="linux_2k.log" sourcetype="Linux logs" ("Failed password" OR "authentication failure" OR "invalid user" OR "user unknown") "user=test"`. The results show 4 events from July 8, 2023, at 20:14:56 PM. Each event is an authentication failure for the user 'test' from the host 'ASUS' via 'Linux_2klog'.

i	Time	Event
>	7/8/26 8:14:56.000 PM	Jul 8 20:14:56 combo sshd(pam_unix)[20964]: authentication failure; logname=uid=0 euid=0 tty=NODEVssh ruser= rhost=212.0.132.20 user=test host = ASUS source = Linux_2klog sourcetype = Linux logs
>	7/8/26 8:14:56.000 PM	Jul 8 20:14:56 combo sshd(pam_unix)[20968]: authentication failure; logname=uid=0 euid=0 tty=NODEVssh ruser= rhost=212.0.132.20 user=test host = ASUS source = Linux_2klog sourcetype = Linux logs
>	7/8/26 8:14:56.000 PM	Jul 8 20:14:56 combo sshd(pam_unix)[20969]: authentication failure; logname=uid=0 euid=0 tty=NODEVssh ruser= rhost=212.0.132.20 user=test host = ASUS source = Linux_2klog sourcetype = Linux logs
>	7/8/26 8:14:55.000 PM	Jul 8 20:14:55 combo sshd(pam_unix)[20963]: authentication failure; logname=uid=0 euid=0 tty=NODEVssh ruser= rhost=212.0.132.20 user=test host = ASUS source = Linux_2klog sourcetype = Linux logs

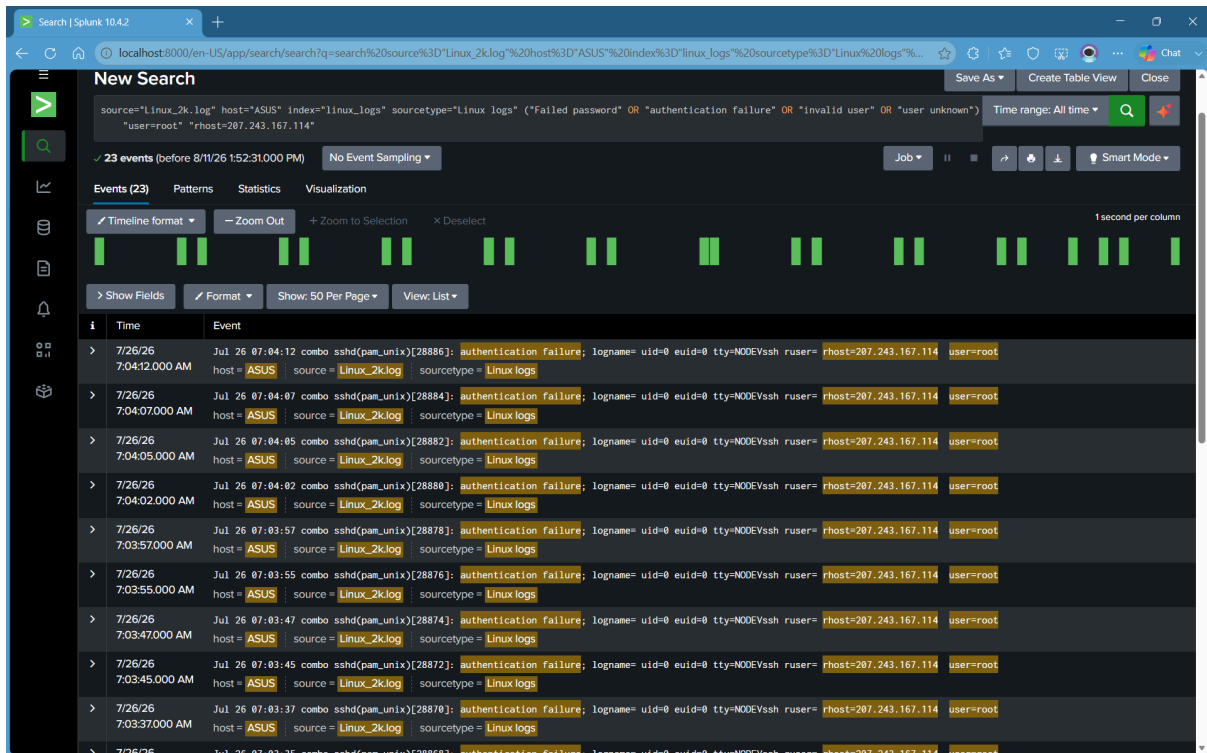
"user=test" resulted in 4 events from src/ip 212.0.132.20 between Jul 8 20:14:55 and Jul 8 20:14:56



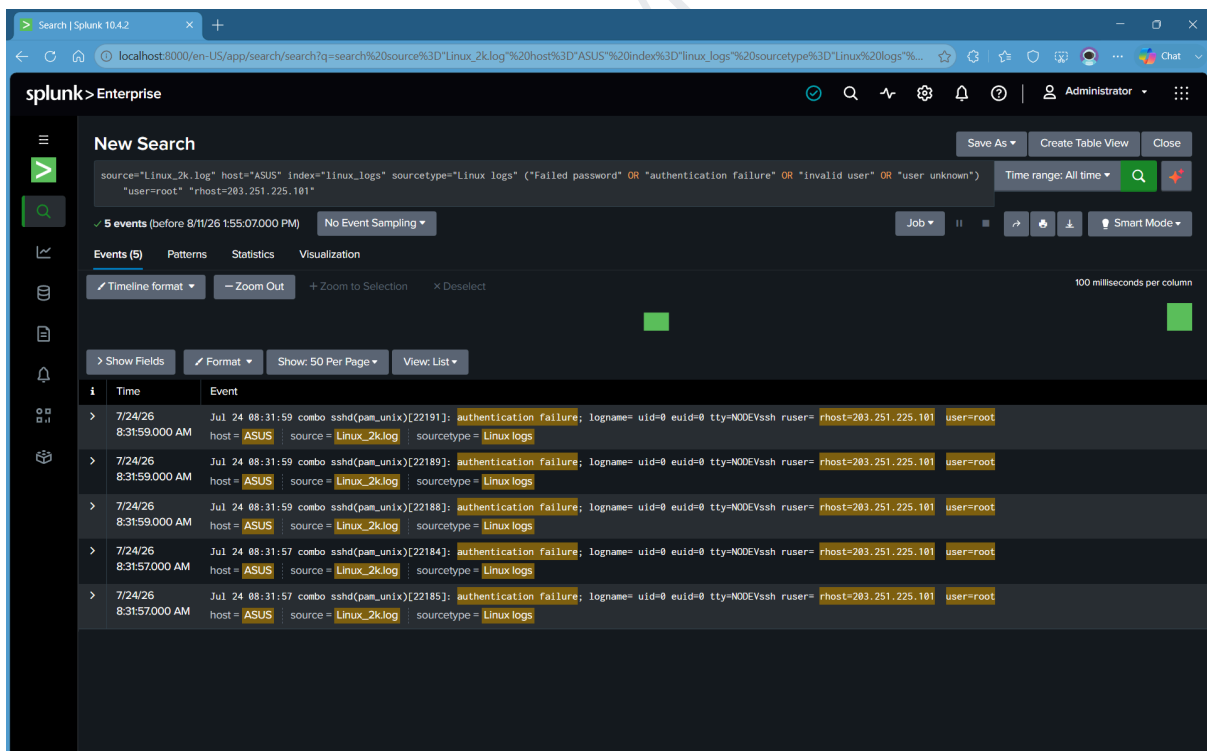
Additionally, including the keyword “user=root” resulted in 351 events



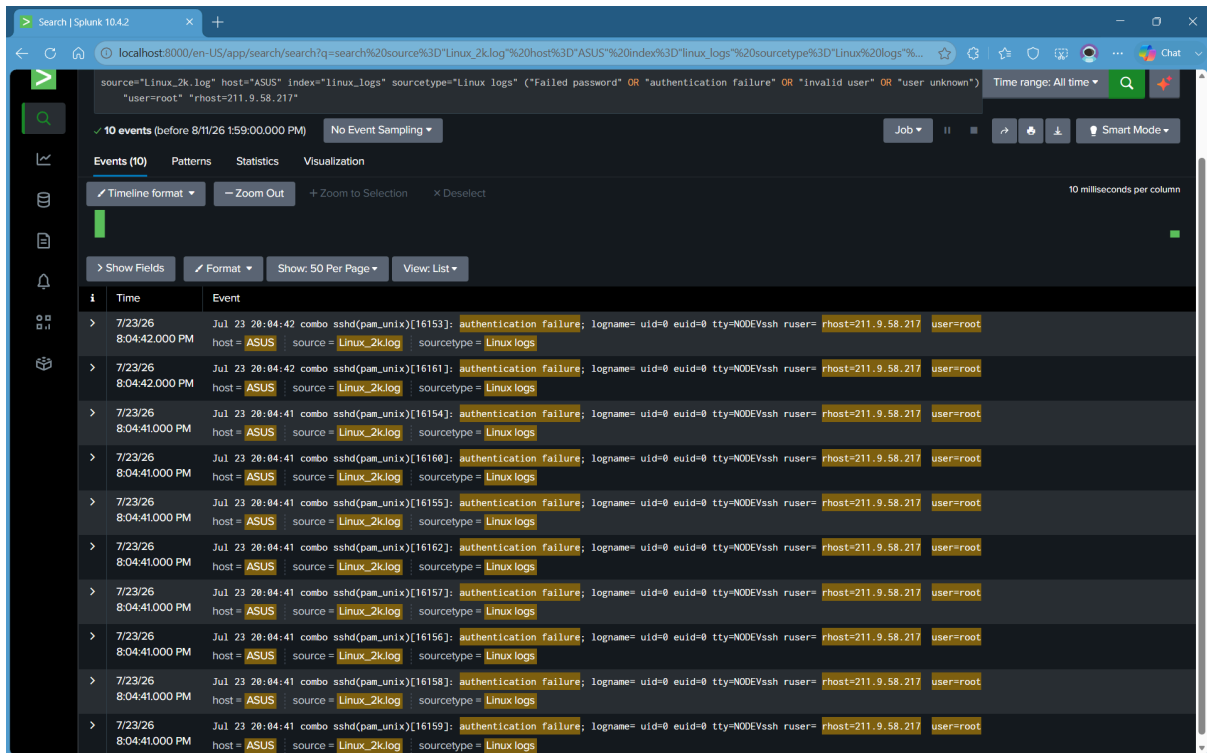
Example 1: src/ip 150.183.249.110 trying to access root user 80 times between Jul 10 16:01:43 and Jul 10 16:03:18



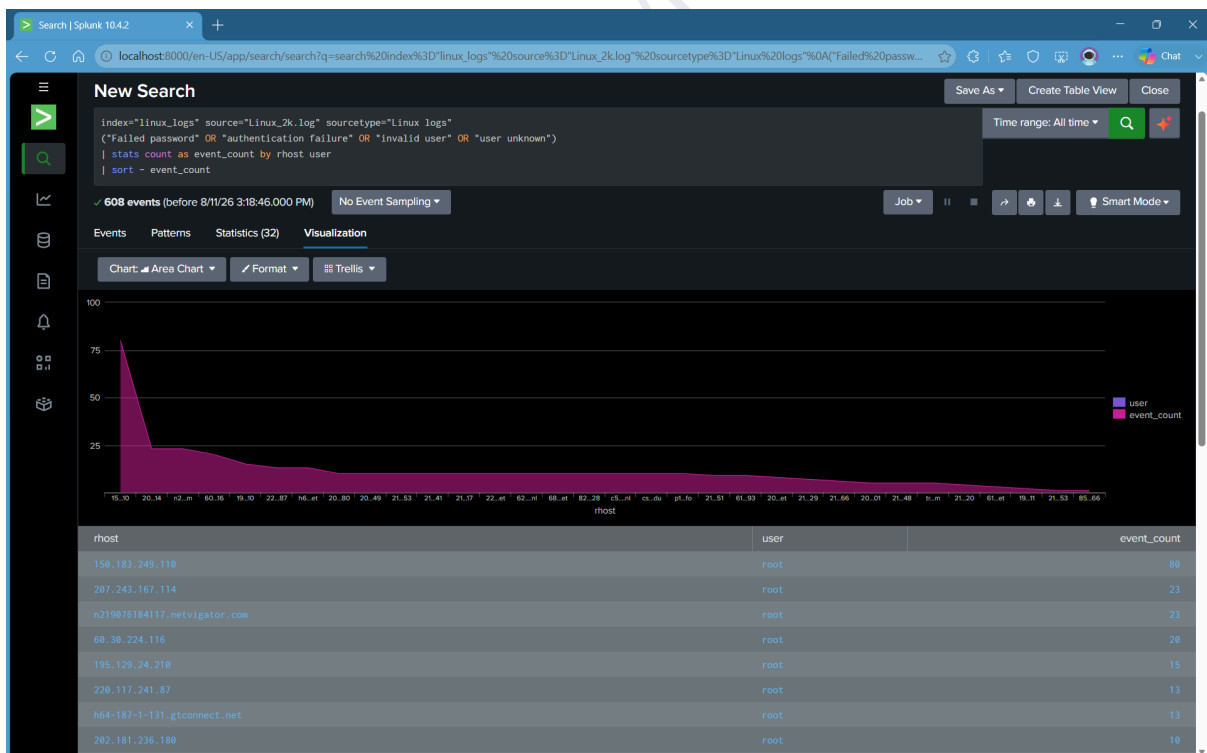
Example 2: src/ip 207.243.167.114 trying to access root user 23 times between Jul 26 07:02:27 and Jul 26 07:04:12



Example 3: src/ip 203.251.225.101 trying to access root user 5 times between Jul 24 08:31:57 and Jul 24 08:31:59



Example 4: src/ip 211.9.58.217 trying to access root user 10 times between Jul 23 20:04:41 and Jul 23 20:04:42



index="linux_logs" source="Linux_2k.log" sourcetype="Linux logs"
 ("Failed password" OR "authentication failure" OR "invalid user" OR "user unknown")
 | stats count as event_count by rhost user
 | sort - event_count

Visual shows 80 attempts at root access from src/ip 150.183.249.110, followed by src/ip 207.243.167.114 with 23 attempts - even though they were days apart, all attempts occurred within short intervals. These suggest automated brute-force attempts using scripts or bots, with focus on gaining administrative access.

Review of Findings

Manual, Python, and Splunk analysis consistently identified repeated suspicious SSH authentication attempts. Manual analysis provided initial context, Python identified 607 suspicious entries across the 2,000-line log, and Splunk identified 607 events and revealed patterns such as repeated attempts against the root account from specific remote IPs.

Reflection

The three approaches complement each other by providing different levels of analysis. Manual analysis helps understand individual log events, Python automation efficiently processes large datasets, and Splunk allows events to be grouped, correlated, and visualised. Together, they provide a more efficient and reliable cybersecurity investigation workflow than relying on a single method.